The Enzyme-Constrained Thermodynamic Flux Module

What it computes, what it is fed,

and what has been measured

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Abstract

The flux module is a differentiable layer that maps a perturbed gene representation to a flux vector over yeast-GEM 9.0.2, subject to stoichiometry, an enzyme-capacity bound and a thermodynamic sign constraint. It is neither a Type I nor a Type II virtual instrument as those were defined: its output is a latent physical state rather than a measured phenotype, and the phenotypes are read off that state.

Its distinguishing property is that three of its inputs are meant to be tabulated rather than learned: turnover numbers, Michaelis constants and formation energies, each hash-pinned and attributable to the weights or model version that produced it. That is what would make the layer a reconstruction of a metabolic network rather than a parameterization fitted to one organism.

None of the three tables built in this work is consumed by any measured arm here. Every arm below resolves 4.0 percent measured k_{cat} from the Open Enzyme Database and defaults the rest, and reads yeast-GEM's shipped formation energies, which are a kcal-scale file labeled kilojoules. The resolvers now carry a predicted k_{cat} tier and a merged energy route, and the cells exercising them have completed one epoch, so every result here should be read as testing the module's structure rather than its parameters.

This document consolidates what was scattered across the 026 and 027 notes: the formulation as implemented, the three parameter tables and the sharp distinction between those wired in and those merely built, the ablation registry, and every measured number. Two corrections to earlier claims are recorded rather than quietly folded in. The ChemAxon licensing blocker was said not to apply to the betaxanthin pathway; it does, and it binds on the compounds that matter, since leucodopachrome, betalamic acid and betanidin are all absent from the eQuilibrator compound cache and only the two generic tyrosine precursors resolve. And the reaction-coverage shortfall is not a mapping gap: an audit that recovers every identifier route available gains 0.58 percentage points, because 375 metabolites match a cache record that carries no structure and are counted as resolved while returning no energy.

The training runs that test whether the module helps have returned complete, and the prespecified decision fails on both criteria. Over the full 24 balanced seeds the anchored module scores -0.0196 test Spearman on the pinned 639-gene split against $+0.0269$ for no module at all, a paired gap of -0.0465 ± 0.0135; the excess of that gap over the same contrast with the training labels destroyed is -0.0372, 95 percent interval $[-0.0724, -0.0031]$. The one arm that does not cost anything is `flux_off`, which keeps the stoichiometry and the box but switches the energies off, at $+0.0388$, level with the published Flux Cone Learning baseline.

An audit of the prediction dumps then showed that every one of the 144 cells is a constant predictor: the largest prediction spread in any cell is 0.8 percent of the target spread, the no-module arms sit at the target mean with a training loss of exactly 1.00 from epoch 10, and the flux arms sit at wrong constants as far as -7 from it. Every correlation in the decision is computed from jitter, so the verdict that the module is not shown to help stands only because nothing in the grid learned the target, and the stronger claim that the thermodynamic anchor costs is withdrawn. The same backbone reached 0.434 on this target under the 019 trainer, so the defect is in the 027 training loop, and a trainer that reproduces that number on the no-module arm now precedes every other step.

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1 What the module computes ✗

1.1 Where it sits ✗

The Equivariant Cell Graph Transformer separates its instruments by what they map between. A **Type I** instrument is representation to representation: the perturbation operators T_ψ^{del}, T_ψ^{OE}, T_ψ^{KD} act inside the gene embedding space and hand on H_{pert}. A **Type II** instrument is representation to output: a readout that emits a measured phenotype.

The flux module is neither, and the reason is worth stating precisely rather than settling by convention. By the letter of the taxonomy it is Type I, since its output feeds further computation rather than a loss. But every existing Type I instrument acts *within* the gene index space at dimension d and preserves equivariance over genes, whereas this one leaves gene index space entirely for reaction space $\mathbb{R}^{4131}$ under a fixed, externally supplied stoichiometry, and its parameters are tabulated physical constants rather than learned weights. It is a physically identified decoder, and its output ν is a latent physical state rather than a phenotype: the phenotypes are read *off* it. Figure 1 places it against the rest of the transformer.

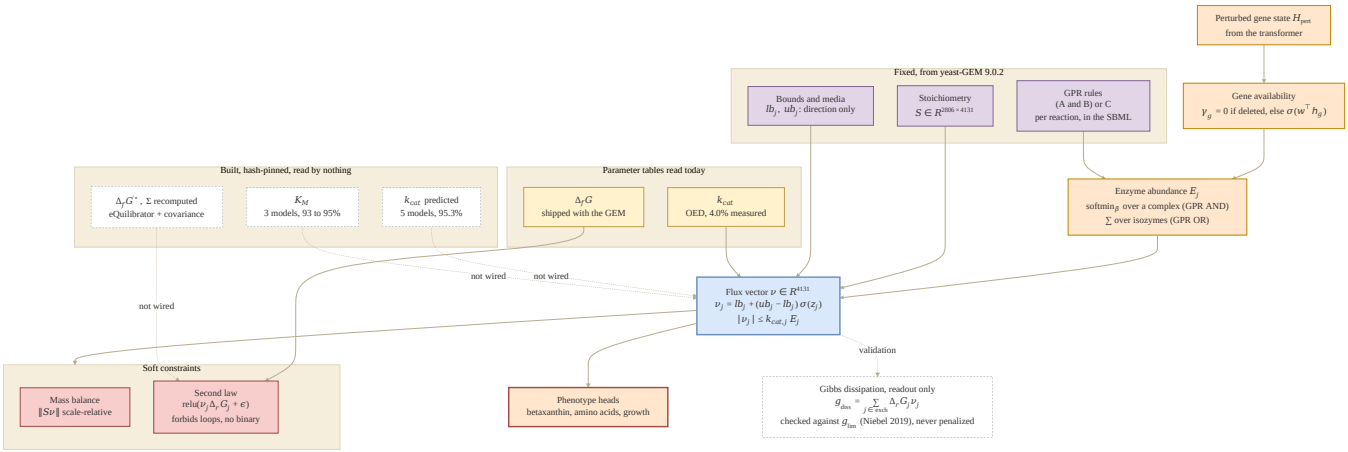


Figure 1: The flux module's main components and what feeds them. Orange is the path from the perturbed gene state to enzyme abundance; purple the fixed yeast-GEM 9.0.2 stoichiometry and bounds; yellow the parameter tables, with a **dashed** border and a dashed edge marking a table that is built and read by nothing; blue the flux vector; red the soft constraints. Every hard constraint is either exact by construction, as the box and the capacity bound are, or relaxed to a smooth penalty; nothing is enforced by a binary variable, which is what makes the layer differentiable. Three of the five parameter tables are drawn dashed because no arm reported here consumes one: k_{cat} resolves from the Open Enzyme Database at 4.0 percent measured and $\Delta_f G$ from the table shipped with the GEM. The predicted k_{cat} tier and the merged energy route now exist in the resolvers and are under test. Section 2 gives the tables and Section 3 their coverage. Same diagram as in the 026 document; source `notes/torchcell.models.equivariant_cell_graph_transformer.mermaid.metabolic-module-simple.md`.

1.2 A program cannot learn, which is the point ✗

The constraint-based tradition solves a linear program per condition. A linear program has no use for a phenotype measurement, so it cannot be improved by one. This layer inverts that: every constraint is either enforced exactly by the parameterization or relaxed to a smooth penalty, so the flux is *conditioned* on roughly 10^7 fitness records, 4,735 betaxanthin measurements and 4,678 amino-acid profiles. **Nothing is enforced by a binary variable**, which is what keeps the whole object differentiable and lets a batch of perturbations cost the same forward pass as a single one.

1.3 The relaxation, term by term ✗

Table 1: Each constraint of an enzyme-constrained thermodynamic model, and how this layer realizes it. "Exact" means the parameterization cannot violate it; "soft" means a penalty that the optimizer trades against the data term.

Constraint	Conventional form	Here
box and directionality	$lb_j \le \nu_j \le ub_j$	**exact**, via sigmoid
enzyme capacity	$\nu_j \le k_{cat} E$	**exact**, folded into the box
mass balance	$S\nu = 0$	soft, scale-relative
second law	$\Delta_r G_j \nu_j < 0$ plus a binary	soft hinge, no binary
Gibbs dissipation	$g_{\text{diss}} \le g_{\text{lim}}$	readout only; hinge available, off by default
protein budget	$\sum_g MW_g E_g \le P_{\text{avail}}$	soft hinge

From representation to gene availability. Each gene's embedding passes through a learned gate, $\gamma_g = \sigma(w^\top h_g)$, so a gene is not merely present or absent. A deletion pins $\gamma_g = 0$; everything else is free to take an intermediate value, which is what leaves room for dosage, hypomorphic alleles and overexpression rather than forcing the layer to treat every perturbation as a knockout.

Folding up the gene-protein-reaction rule. Availability is folded to the reaction through the GPR: a **softmin** over the subunits of a complex, since a complex is limited by its scarcest member, and a **sum** over isozymes, since they substitute for one another. The softmin temperature is a real parameter, not a detail: too soft and a complex behaves like a mean over its subunits, which loses exactly the limiting-subunit behavior the softmin was chosen for.

The box, which is where capacity lives. Flux is parameterized as

$$\nu_j = lb_j + (ub_j - lb_j)\,\sigma(z_j), \qquad |\nu_j| \le k_{cat,j}\,E_{g(j)}, \tag{1}$$

so the bounds and the capacity bound hold by construction and cannot be violated no matter what the network does. Turnover enters in units of h^{-1} ($k_{cat}\,[\mathrm{s}^{-1}] \times 3600$), which is what fixes the units of the enzyme abundance E. The exactness budget is spent here rather than on $S\nu = 0$, and that is a deliberate choice: bounds are exact and mass balance is soft, not the reverse.

The second law, which is the load-bearing relaxation.

$$C_{\mathrm{th}} = \frac{1}{|\mathcal{J}|}\sum_{j\in\mathcal{J}} \mathrm{relu}\big(\nu_j\,\Delta_r G_j + \epsilon\big), \tag{2}$$

with $\mathcal{J}$ excluding exchange reactions, the biomass reaction and water transport. Thermo-Flux exempts the same three classes (Box 2, "Reactions where the second law of thermodynamics is not applied"), and each is exempt for a different reason. An **exchange reaction** has one participant and no chemistry: it is the bookkeeping pseudo-reaction that moves a metabolite across the system boundary, so its "reaction energy" is the formation energy of one compound rather than a difference between reactant and product, and a sign constraint on it would be a constraint on the medium, not on the cell. The **biomass reaction** is a lumped pseudo-reaction whose energy is an empirical estimate from elemental composition (Battley 1999), not a sum over components, and its direction is the objective rather than something thermodynamics should be asked to decide. **Water transport** between compartments is treated as freely reversible: water's activity is fixed at one on both sides by the Alberty convention, the same convention that fixes the proton's energy at zero, so the transport carries no driving force and a hinge on it would penalize noise. Every other reaction, including every other transport process, is inside $\mathcal{J}$.

Gibbs dissipation is a readout, not a constraint. Thermo-Flux introduces the dissipation cap separately from its constraint set, as an option, and its only warrant is Niebel et al.'s regression: a plateau in g_{diss} at $-3.7\,\mathrm{kJ\,gCDW^{-1}h^{-1}}$ above growth rate $0.3\,\mathrm{h}^{-1}$, with the authors' own caveat that a correlated third factor cannot be excluded. It is an empirical limit fitted to one organism's physiology, not a law. The decision here is to keep g_{diss} as a logged quantity and never as a penalty: a model that respects the limit without being told to has learned something about the cell, and a model that violates it has been caught, whereas a model trained to sit under it has been told the answer. Every grid reported in this document ran with the penalty on, since the code default was on until 2026-09-08; the default is now off, and the value stays in the per-epoch log.

The ϵ is not decoration. The conventional big-M pair admits $\nu_j = \Delta_r G_j = 0$ under either value of its binary, so the strict inequality is never actually realized, and a smooth version has to name its tolerance rather than inherit a fiction.

Loop-freedom comes free, and that is the trick. Because $\Delta_r G$ is a difference of potentials it sums to zero around any cycle, so no cycle can run downhill at every step. Requiring $\nu_j \Delta_r G_j \le 0$ therefore forbids internal loops with no integer variable at all, which is precisely the property that makes loopless flux balance analysis a mixed-integer program.

1.4 Free against anchored thermodynamics ✗

The in-cell reaction energy decomposes into three terms,

$$\Delta_r G_j = \underbrace{\Delta_r G'^{\circ}_j}_{\text{tabulated}} + \underbrace{RT\sum_{i\neq \mathrm{H}^+} S_{ij}\ln c_i}_{\text{concentration}} + \underbrace{\Delta_r G^t_j}_{\text{transport}}\,. \tag{3}$$

The distinction between the two thermodynamic arms is exactly which of these is supplied. In the **free** arm the potential is learned, so the layer gets loop-freedom and nothing else; the constraint is structural. In the **anchored** arm

the standard term is read from the tabulated formation energies, so the sign of every reaction is pinned by measured chemistry rather than by whatever the optimizer finds convenient. That difference is the experiment: structural loop-freedom is a real constraint on its own, and the question is whether the chemistry adds anything to it.

1.5 Why enzyme constraints are not optional ✗

The strongest argument for the layer is a property of the model it wraps. yeast-GEM's `listOfParameters` holds **five distinct bound values**: -1000 on 1,667 lower bounds, 0 on 2,462 lower and 7 upper, $+1000$ on 4,123 upper, -1 on exactly one lower bound (D-glucose exchange), and 0.7 on both bounds of the non-growth-associated maintenance reaction. There is no $\pm\infty$ anywhere.

So **4,129 of 4,131 reactions carry bounds that encode direction and on-off and no capacity whatsoever.** The one fitted physiological constant is the maintenance flux; the glucose bound of -1 is an arbitrary normalization, and growth scales with it almost linearly, $0.0858\,\mathrm{h}^{-1}$ at -1 against 0.8877 at -10 and 1.7786 at -20. The whole medium is sixteen numbers.

A network specified that loosely has an enormous feasible set, and the two things that narrow it are the two the layer supplies. Requiring growth at 90 percent of optimum narrows 2,316 reactions, 56.1 percent of the network, and improves median flux-variability width ninefold from 1.077 to 0.1176. **It does not touch the loops**: reactions with width above 10^3 go from 428 to 427. That residue is what the thermodynamic term exists to remove, and it is why loop-freedom rather than capacity is the load-bearing constraint.

2 The three parameter tables ✗

2.1 What was there before, and what is there now ✗

The module needs three tables and had almost none of them. Measured k_{cat} covered 148 of 3,728 catalytic units, 4.0 percent, with the other 96 percent running on a single flat default of $10.3\,\mathrm{s}^{-1}$. A catalytic unit here is a **(reaction, AND-term) pair**, not a distinct enzyme complex: `parse_catalytic_units` assigns an id inside the per-reaction loop, so one complex catalyzing five reactions counts five times. The deduplicated AND-term count is 1,065. Both numerator and denominator are counted the same way, so 4.0 percent is internally consistent, but the denominator has to be named precisely or the reader computes 148/1,065 and gets 13.9. K_M had no table at all: the parameter existed as an enum member and a protocol flag with nothing behind either. The thermodynamics came from a shipped file of point values with no covariance, usable only for this organism.

Seven kinetic predictors were mirrored into `$DATA_ROOT/models/kinetics/<name>/` with per-file `sha256`, retrieval commands and git commits, and all seven now run over the enzyme-substrate pairs of yeast-GEM 9.0.2, with protein sequences taken from the genome object rather than from the table each model bundles. The seventh, EZKinetics, is unpublished and pinned by repository and commit (`le-yuan/EZKinetics`, revision `bf7a897`); it is the only one that conditions on assay pH and temperature, and since yeast-GEM carries neither per pair it is run at the authors' 7.0 and 25 °C defaults, which keeps it comparable with the six condition-blind predictors.

Table 2: Every parameter table now built, with the exponentiation base established by validating each predictor against its own authors' published output. Coverage is of the GEM's 3,728 catalytic units.

Predictor	Parameter	Rows	Units	Coverage	Median
DLKcat	k_{cat}	7,351	3,552	95.3%	4.012 s^{-1}
UniKP	k_{cat}	7,351	3,552	95.3%	4.050 s^{-1}
EITLEM	k_{cat}	7,456	3,552	95.3%	3.981 s^{-1}
TurNuP	k_{cat}	5,143	2,056	55.2%	10.819 s^{-1}
DeepEnzyme	k_{cat}	7,300	3,546	95.1%	4.167 s^{-1}
EZKinetics	k_{cat}	7,456	3,552	95.3%	4.617 s^{-1}
UniKP	K_M	7,351	3,552	95.3%	0.0796 mM
EITLEM	K_M	7,456	3,552	95.3%	0.0793 mM
Boost_KM	K_M	7,157	3,468	93.0%	0.1410 mM
EZKinetics	K_M	7,456	3,552	95.3%	0.1424 mM

Two of these models exponentiate base 2 and five base 10. DLKcat and DeepEnzyme are $\log_2$; UniKP, EITLEM, TurNuP, Boost_KM and EZKinetics are $\log_{10}$. Reading either wrong yields turnover numbers that still look entirely credible, which is why each was validated against something its own authors published before its values were believed.

2.2 Which tables are actually wired in ✗

This distinction decides how any result from the module should be read, and it is easy to lose.

Table 3: What the layer consumes today. A table that is built but not consumed contributes nothing to a measured result and must not be described as if it did.

Table	Status	How it enters, or why it does not
k_{cat}	**wired**; the predicted tier exists and no arm below used it	the capacity bound $\|\nu_j\| \le k_{cat,j} E_{g(j)}$. Every arm in this document ran `resolve_kcat_table` on the OED mirror alone, 4.0 percent measured and a default for the rest. A `PredictedUnitTable` tier now sits between the measured values and the default, filling 3,410 of the 3,728 units, and is exercised only by the wave-5 cells, which are still running
$\Delta_f G'^{\circ}$	**wired**; the merged route exists and no arm below used it	the standard term of $\Delta_r G_j$, hence the second-law hinge and the dissipation bound. Every arm below read `data/databases/model_metDeltaG.csv` through `load_thermo_table`, which is a kcal-scale file and was read as kilojoules, so those hinges were 4.184 times too permissive; the reader converts at parse time now. `resolve_reaction_delta_g` builds from a named route, and the union of the shipped and recomputed metabolite tables prices 3,722 reactions with a propagated standard deviation on 1,755
K_M	**built, not wired**	the saturation term is drawn as inactive in the module diagram and is inactive in the code. K_M contributes nothing to any result reported here

None of the three tables built in this work is consumed by any measured arm below. Every arm reported here ran on the Open Enzyme Database's 4.0 percent measured k_{cat} plus a default, and on yeast-GEM's shipped formation energies. That is the sharpest thing to carry out of this document, because the module diagram draws solid edges from the kcat and thermodynamic tables into the box and the potential, and no arm below has those edges.

The plumbing itself now exists and is under test. `resolve_kcat_table` takes a predicted tier and `resolve_reaction_delta_g` a named energy route, and GilaHyper cells are running the flux arms against them under a constraint warmup: job 1663 (`flux_off`), jobs 1667 and 1668 (`flux_capacity` and `flux_abundance` on the unit-corrected union, resubmitted after jobs 1664 and 1665 were found to be reading a mixed-unit table) and job 1669 (head normalization). None has reached a selected epoch or a test score, so nothing here is a result for the predicted or recomputed tables. K_M still has no consumer at all.

2.3 Why six predictors and not one ✗

Re-keying the comparison from the enzyme-substrate pair to the **gene**, which is the unit the capacity bound reads, gives the number that decides how a predicted table should be used.

Table 4: Across-predictor spread per gene against the dynamic range of the quantity itself, in orders of magnitude, over genes every predictor covers. The k_{cat} row is over six predictors and the K_M row over four; before EZKinetics they were 1.23 against 0.94 (ratio 1.32) and 0.70 against 1.34 (ratio 0.52), so the sixth predictor widened both disagreements.

Quantity	Genes	Median spread	Consensus p10–p90	Ratio
k_{cat}	980	**1.28**	0.87	**1.47**
K_M	1,150	0.81	1.22	0.67

For k_{cat} the disagreement between predictors exceeds the quantity being predicted, and a sixth predictor widened it. The median gene moves 1.28 orders of magnitude between the highest and lowest predictor, a factor of 19, and 71 percent of genes move by more than a full order of magnitude, against a 10th-to-90th-percentile spread across genes of 0.87 orders of magnitude. Choosing a different published predictor moves one gene's turnover number further than real enzymes differ from one another. Gene-level Spearman between k_{cat} predictors runs 0.01 (DeepEnzyme against EZKinetics) to 0.55 (EZKinetics against UniKP).

For K_M it does not. The same construction gives 0.81 against 1.22, and Boost_KM agrees with UniKP at a gene-level Spearman of 0.75, higher than any pair of k_{cat} predictors reaches. Both are statements about agreement rather than accuracy, since almost none of these pairs has a measured value.

That is why the registry in `torchcell/datasets/kinetics/` exists and why a config string selects the predictor: predictor choice is a hyperparameter with a measured effect size, not a settled detail.

2.4 Which predictor is right, as far as that can be measured ✗

Agreement is silent on accuracy. The Open Enzyme Database mirror carries measured *S. cerevisiae* values, and restricting to wild-type enzymes assayed between 20 and 40 °C and joining on (UniProt accession, substrate) leaves **57** pairs that are also among the catalytic units the predictors were run over.

Table 5: Accuracy on the 57 measured pairs. Error is the median absolute $\log_{10}$ ratio in orders of magnitude, bias the median signed ratio, negative meaning the predictor runs low. Every predictor trained on BRENDA or SABIO-RK, from which the Open Enzyme Database is itself assembled, so an unknown fraction of these pairs was seen in training and the values are an upper bound on held-out accuracy.

Parameter	Predictor	Spearman	Error	Bias	Within 1 dec
k_{cat}	EZKinetics	**+0.790**	0.37	−0.14	**91.2%**
k_{cat}	UniKP	+0.751	0.31	−0.12	89.5%
k_{cat}	DLKcat	+0.656	0.38	−0.14	84.2%
k_{cat}	EITLEM	+0.603	**0.23**	−0.01	78.9%
k_{cat}	TurNuP	+0.506	0.44	+0.00	81.1%
k_{cat}	DeepEnzyme	+0.341	0.65	−0.23	66.7%
K_M	EZKinetics	**+0.701**	0.40	−0.05	**87.7%**
K_M	Boost_KM	+0.680	0.53	−0.16	78.9%
K_M	UniKP	+0.615	0.61	−0.08	78.9%
K_M	EITLEM	+0.612	**0.43**	+0.02	82.5%

The accuracy ranking is not the agreement ranking. EZKinetics leads on rank accuracy for both parameters while agreeing with DeepEnzyme at 0.01 on k_{cat}; EITLEM leads on magnitude and places fourth on rank. So Table 4 cannot be read as a quality ordering, and the two accuracy axes disagree about which predictor is best. The 57 scored pairs are not the easy tail either: the across-predictor k_{cat} spread on them is 1.37 orders of magnitude against 1.27 over all 3,814 pairs, so the upper-bound caveat rests on leakage alone. If one predicted table were to be wired into the capacity bound, EZKinetics is the one these numbers point at for both parameters, with a k_{cat} bias of −0.14 orders of magnitude to correct.

Most of that correlation is the enzyme class rather than the enzyme, and this bounds what wiring a predicted table can buy. Measured on the whole Open Enzyme Database, 35,753 records across 4,555 enzymes, the median EC4 class spans 1.77 orders of magnitude between its highest and lowest member against a measurement floor of 0.12 orders of magnitude, so knowing an enzyme's exact four-digit EC number leaves 0.75 orders of magnitude of standard deviation unexplained and accounts for under half the variance (intraclass correlation 0.46). A lookup table that predicts each enzyme as the median of the others in its EC class, using no sequence and no learning, scores a Spearman of **0.430** on the same 57 pairs: below TurNuP and above DeepEnzyme. Centering both axes on the class mean collapses every predictor's Pearson from the 0.81 to 0.93 range to between −0.16 and +0.53, and only EZKinetics has an interval excluding zero.

The consequence for this module is specific. A flux layer already knows which reaction it is parameterizing, so the between-class range is information it does not need; what the capacity bound needs is the within-class ordering, which is the part these predictors are measured to be weakest at. A predicted k_{cat} table is therefore an order-of-magnitude prior, and expecting the wiring to sharpen the bound is not supported by the accuracy numbers. The full analysis, with its sample-size caveats, is in `notes-tex/026-kinetics-thermodynamics`, Section 4.5.

3 Thermodynamic coverage, and two corrections ✗

3.1 The recomputation ✗

Formation energies were recomputed from eQuilibrator's component contribution, which returns a covariance and applies to any organism because it works from structure rather than from a curated per-organism file. Summing the resulting table reproduces eQuilibrator's own reaction call to 6.8×10^{-13} kJ mol^{-1}, so the table is arithmetically correct. On the chemical reactions both tables cover it agrees with the shipped file to a median of 16.25 kJ mol^{-1}, which is worse than the shipped file agrees with itself, and is expected once that file is read for what it is: yeast-GEM's metabolite and reaction columns come from three sources (yETFL, ModelSEED, dG-predictor) under three different matching keys, one of them row position.

The shipped columns are also in kilocalories and labeled kilojoules. They are yETFL's values, from Oftadeh et al. 2021, copied by yeast-GEM's own curation script with no conversion, and yETFL carries `thermo_units = 'kcal/mol'`; regressing our kJ mol^{-1} formation energies on yETFL's over 1,927 shared metabolites gives a slope of 4.187 at Pearson $r = 0.987$. The 33 reactions taken from dG-predictor are in kilojoules, so the reaction column mixes

two unit systems. Correcting the unit and matching the compartment conditions closes the *transport* arm of the disagreement, median 17.4 to 5.6 kJ mol^{-1} at r 0.30 to 0.75, and leaves the single-compartment cloud where it was, 16.25 to 17.60. Only the second is a disagreement about thermodynamics. Every arm below read that column as kilojoules; `_read_delta_g_csv` now multiplies by 4.184 at parse time, so no later code path can hold a kcal number and call it kJ. The 33 reaction-column values that came from dG-predictor are already kJ and are over-converted by that factor; they supply 6 of 4,131 reactions in the union's last-resort fill and nothing else.

The gain is 785 reactions with a real standard error, median 2.30 kJ mol^{-1}, and the rank-590 covariance factor the layer had recorded as unavailable. The cost is reaction coverage: 66.4 percent against the shipped file's 84.9.

3.2 Correction: most of the shortfall was the proton ✗

Reaction coverage was 42.5 percent, and that number was read as the price of an organism-independent method. Ranking metabolites by how many otherwise-priceable reactions each blocks says otherwise: the top twelve blockers were all H^+, in twelve compartments, blocking 1,477 reaction slots between them.

The proton is the reference of the Legendre transform, not a compound to be priced by it. `standard_dg_formation` returns nothing for it, correctly, because a bare proton has no group decomposition. Left as a missing value it deleted every reaction that moves a proton. Setting it to zero, which reproduces eQuilibrator's own reaction call to 0.0000 kJ mol^{-1} on four proton-carrying balanced reactions, takes coverage from 1,755 reactions to **2,741**, a gain of 986. It does not move the union with the shipped table, which stays at 3,716: every added reaction was already covered, so what improved is independent confirmation rather than coverage.

The fix carries a known error that is worth more attention than the gain. Zero is exact within one compartment and wrong across a membrane, where $RT\ln(10)\,\Delta\mathrm{pH}$ per proton is the entire driving force. **183 reactions move a proton between compartments and 170 are now priced**, each missing that term: previously unpriced, now confidently wrong.

That term now exists as a separate table. `torchcell/metabolism/transport.py` evaluates $\Delta_r G_j^t = -N_H RT\ln(10)\,\Delta\mathrm{pH} - Fq\,\Delta\Phi$ per reaction from the stoichiometry, each metabolite's compartment and its charge, giving a nonzero term on 326 of the 1,468 reactions that cross a membrane. Against yeast-GEM's own column, converted out of kcal, the sign agrees on 84.7 percent of the 287 reactions nonzero in both at Pearson $r = 0.804$ (`results/transport_terms.json`). It is a configuration field, `transport_delta_g_source`, and it defaults to `off`, so every arm reported in this document still carries the zero. The 613 reactions with a nonzero shipped term and a zero computed one are the protonation contribution to N_H, which needs a species distribution per transported compound and is not computed.

3.3 The remaining shortfall is not a mapping gap ✗

The next reading is that identifiers are failing to resolve, and that better mapping would close what is left. An audit that implements every available recovery route measures otherwise.

There are two loss channels, and the summary reported only one. The 332 "unresolved" metabolites are the accession-match failures. A further **361 metabolites match a cache record that carries no structure and no group vector**, so `standard_dg_formation` returns nothing while the tier histogram counts them as resolved. Total unpriced is **693**, not 332. The baseline is 2,113 of 2,806 metabolites, 75.30 percent, and 2,741 of 4,131 reactions, 66.35 percent.

Table 6: Every identifier route available, applied in turn. Reaction coverage is the binding number because a reaction needs every participant priced.

Step	Metabolites priced	Reactions	Δ reactions
baseline	2,113 (75.30%)	2,741 (66.35%)	–
extra identifier namespaces	2,116	2,753 (66.64%)	+12
MetaNetX cross-reference table	2,127	2,764 (66.91%)	+11
InChIKey structure lookup	2,141 (76.30%)	2,787 (67.47%)	+23

Total recovery is 28 metabolites and 46 reactions, 1.12 percentage points. The extra namespaces are worth exactly one chemical, hydrogen sulfide. The census explains why: the model carries chebi on 2,402 metabolites and metanetx on 2,251, but biocyc on 29, seed on 16, reactome on 2, and no hmdb, lipidmaps or swisslipid at all. Of the 332 unmatched, **181 of the 192 distinct chemicals carry no chemical annotation whatsoever**, and the class breakdown is overwhelmingly lipid: phosphatidylinositol 135, cardiolipin 62, triacylglycerol 34, phosphatidate 25.

Of the 665 still unpriced, 235 (35.3 percent) are a real ceiling: 185 carry a placeholder formula with an R, X or $*$ so no definite structure exists, and 50 are known to eQuilibrator but have no group decomposition. The other 430 (64.7 percent) have a definite formula and no usable structure anywhere, which is a structure-curation gap rather than a limit of the method.

Most of that gap is now closed. 409 of the 430 are lipids whose acyl composition is written into the name, and `lipid_structure_writer.py` parses 168 of the 206 distinct names, writes a structure for each, and reproduces yeast-GEM's own formula on all 168. **347 of the 409 metabolite rows are priced**, 289 by finding a compound eQuilibrator already holds and 58 through the external pK_a route (`results/lipid_coverage.json`). The eQuilibrator recomputation moves from 66.35 to 86.27 percent of reactions and passes the shipped column's 84.85, while the union with the shipped table does not move at all, since that table already prices these lipids by name. The gain is method consistency rather than coverage. Two conditions hold the values out of the built table for now: they are all at pH 7.2 and need re-transforming per compartment, and the 48 names priced only through the external route sit 777.5 kJ mol^{-1} off a cache value because they were written as the neutral parent rather than the major microspecies, a fix that is in progress.

3.4 A correctness bug in the shipped build, found by the same guard ✗

The InChIKey route initially reported a gain six times larger, and it was wrong. MetaNetX writes `[*]` dummy atoms for polymers and conjugates; RDKit returns an *empty* InChIKey for these rather than failing; an empty key reaches the cache as `LIKE '%'` and matches the entire compound table. `Ala-tRNA(Ala)` was being priced as water.

The guard that fixes it, comparing heavy-atom composition against the GEM formula with hydrogen excluded because protonation state legitimately differs, then finds the same class of error in the build already in use: **31 metabolites currently carry a formation energy from a compound whose heavy-atom composition disagrees with the GEM formula**, and 14 of the covered reactions touch one. Every dolichol species is matched to a C25 pentaprenyl against the GEM's C20 tetraprenyl; `glycogen` is matched to a C24 tetramer against a C6 monomer; `protein`, `carbohydrate` and `sterols` are matched to concrete molecules.

This does not raise coverage and it is worth more than coverage: it removes 31 wrong numbers. It is not yet applied to `resolve_compound`, and it should be.

3.5 Where the ChemAxon license actually binds ✗

eQuilibrator's component contribution needs a pKa and majority-microspecies decomposition for any compound it has never seen, and it obtains that by shelling out to ChemAxon's `cxcalc`. A compound already in the compound cache ships with that decomposition precomputed, so no license is involved. The license binds only on compound *creation*.

Two corrections are recorded here rather than folded away. An earlier account of this work stated that the blocker does not apply to betaxanthin because all of the pathway's intermediates are in the cache. A probe on 2026-09-04 then reported three of the five as absent. The probe was wrong for a reason of its own: it matched the full InChI string, and the cache stores the InChI of the major microspecies at pH 7, whose final InChIKey block records the protonation state. Matching on the first two InChIKey blocks, the skeleton and stereochemistry, finds all five, each with exactly one cache compound, a group vector and ChemAxon pKas. The first account was right in its conclusion and wrong in its evidence; the probe was wrong in both.

Table 7: Every compound of the DeLoache pathway against the eQuilibrator cache, resolved by InChIKey prefix. The formula of all five matches PubChem. The three pathway-specific compounds resolve to the cache's deprotonated microspecies, which is why an exact-InChI match missed them. Energies in kJ mol^{-1} at pH 7.2, with the component-contribution standard deviation.

Compound	Role	$\Delta_f G'^{\circ}$	σ	Status
L-DOPA	CYP76AD1 hydroxylation product	−157.7	4.30	in cache
dopaquinone	CYP76AD1 oxidase side activity	−168.3	7.54	in cache
leucodopachrome	cyclo-DOPA, spontaneous cyclization	−233.0	8.79	in cache
betalamic acid	DOD product, the reactive aldehyde	−366.0	8.49	in cache
betanidin	violet pigment, competing branch	−394.9	8.56	in cache

All five are priced, so the license binds nowhere on this pathway. A thermodynamic statement about the betaxanthin route is available from the cache alone, with the three pathway-specific compounds carrying a standard deviation near 8.5 kJ mol^{-1} against 4.3 to 7.5 for the two native precursors.

This does not affect any number reported elsewhere in this document, because the flux layer reads yeast-GEM's shipped table and the pathway is heterologous, hence absent from the GEM entirely.

For a genuinely novel compound the constraint has also loosened. Upstream `equilibrator-assets` merged support on 2026-08-15 for creating compounds without ChemAxon when pKa values are supplied, via `pkas_provided_externally` and `specified_pkas`. That is master-only; the pinned release 0.6.0 does not contain it. What the interface needs is an unannotated list of macroscopic pKa floats, which is a weaker requirement than the site-annotated micro-pKa the harder tools produce. That route was benchmarked on 2026-09-10 against 300 cache compounds whose stored ChemAxon constants serve as ground truth: Uni-pKa on its simple enumeration template (Luo et al., *JACS Au* 4:3451–3465, 2024), run locally on Apache-2.0 weights, moves the pH 7.2 formation energy by 0.91 kJ mol^{-1} mean absolute error, median 0.16 and 90th percentile 2.84, well inside the 4.3 to 8.8 kJ mol^{-1} uncertainty eQuilibrator itself attaches to the betaxanthin compounds. MolGpKa reads 1.21 and pKasolver-lite 2.52. So the license is replaceable. What the route cannot do on its own is enlarge the model: only 16 of the 665 unpriced yeast-GEM metabolites lack nothing but pKas, and the 409 that could follow needed the lipid structure writer above. With it, the measured reach is 347 of 665, 52.2 percent. The benchmark, the bridge audit and the reach analysis are in the notes `experiments.026-metabolism-flux.external-pka-route` and `experiments.026-metabolism-flux.lipid-structure-writer`.

4 The ablation registry ✗

4.1 What each arm turns on ✗

The registry is one dictionary, `ARMS` in `experiments/026-metabolism-flux/scripts/train_flux.py`, and it is the single source of truth for what the layer does in any run. An experiment names an arm; it never redefines one. That is what keeps a difference between two arms attributable to the module rather than to a hyperparameter that moved alongside it.

Table 8: The registry as implemented. Each row is a strict addition to the one above it except `nullspace`, which changes the parameterization rather than the physics.

Arm	Flux	Thermo	Capacity, budget, dissipation	Isolates
`pooled`	off	–	–	the no-module control
`flux_off`	on	`OFF`	off	routing through S alone
`flux_free`	on	`FREE`	off	structural loop-freedom
`flux_anchored`	on	`ANCHORED`	on	**the treatment**
`flux_nullspace`	on	`ANCHORED`	on	the exactness budget

The ladder is deliberate. `flux_off` routes the representation through the stoichiometric network with nothing but mass balance and the GPR, so it asks whether the network's topology alone carries signal. `flux_free` adds a learned potential, which buys loop-freedom and no chemistry. `flux_anchored` replaces the learned potential with tabulated formation energies and switches on the enzyme capacity bound, the protein budget and the dissipation limit. `flux_nullspace` keeps the anchored physics but spends the exactness budget on $S\nu = 0$ instead of on the box, which is the one arm that tests the parameterization choice rather than the model.

4.2 The control that is missing, and why it matters ✗

Every arm above changes the module's *physics* together with its *parameter count*. So a gap between `flux_anchored` and `pooled` is consistent with two different explanations: the tabulated chemistry is informative, or the extra structure acts as a capacity constraint that would help with any numbers in it.

The arm that separates those is `flux_shuffled`: the full anchored module with k_{cat} **permuted across reactions**, holding architecture, parameter count and optimization fixed and destroying only the correspondence between a reaction and its turnover number. If `flux_anchored` and `flux_shuffled` land together, the kinetic tables are decoration.

It is **not in the registry and not in the current run.** Adding it is a code change that was not worth making untested, and it is the first item of the next round.

4.3 The permutation nulls ✗

Two arms exist only to be compared against: `null_pooled` and `null_anchored` train on **permuted labels** while validating on real ones. They matter because of what 026 measured. Against zero, its flux-versus-pooled gap of +0.0735 is 7.7 standard errors. Against its own permuted-label gap of +0.0302, the excess is +0.0433 at 1.9 standard errors. More than 40 percent of "the flux module helps" survived destroying the labels.

So the decision rule is a difference of differences,

$$\big(\text{flux} - \text{pooled}\big) - \big(\text{null flux} - \text{null pooled}\big), \tag{4}$$

and a gap measured against zero is not evidence.

5 What has been measured ✗

5.1 What separates the two rounds ✗

Both rounds ask whether the flux module helps predict betaxanthin, and they differ in almost everything else. Reading a number from one against a number from the other is the mistake this table exists to prevent.

Table 9: The two rounds, side by side. The backbone is the difference that outranks the others, since the 026 baseline was roughly five times below the demonstrated ceiling on this target.

	026	027
backbone	weak: 2 graphs, learnable embeddings, hidden 32, 4 heads	strong: 9 graphs, `prot_T5_all`, no learnable table, hidden 90, 9 heads
trainer	`sweep_flux.py`	`train_bx.py`
epochs	30	200
split	re-rolled per seed, so seed moves signal and nuisance together	**pinned** Merzbacher 639-gene test split, fixed across every cell
scored on	validation, **maximum over epochs**	test, at an epoch selected on validation
replication	10 seeds per arm, plus a 24-run permuted-label null	24 seeds per arm, with the nulls as two of the six arms
machine	GilaHyper	Delta `gpuA40x4`, 72 workers
W&B project	`torchcell_026_flux_sweep`	`torchcell_027_bxfx`
outcome	no arm clears its own null	every cell is a constant predictor

5.2 The 026 sweep, and why it decided nothing ✗

Table 10: The 026 sweep on betaxanthin. The statistic is validation Pearson scored as a *maximum over epochs*, which is an upward-biased order statistic, and the last column is the mean of the final five epochs, which is not.

Arm	Seeds	Peak mean	s.e.m.	Peak epoch	Last-5 mean
`flux_anchored`	10	0.1151	0.0095	17	0.0057
`flux_free`	10	0.1020	0.0115	12	0.0212
`flux_off`	10	0.0837	0.0112	10	0.0044
`pooled`	10	0.0416	0.0098	4	−0.0041
`flux_nullspace`	10	0.0381	0.0114	0	0.0000

Every run behind that table has a W&B page, indexed in Table 11; the arm rows there are the ten seeds each arm contributes, and the null and regularization grids are the runs the next two paragraphs draw on.

Table 11: Every run of the 026 sweep behind Table 10, 98 in all, in project `torchcell_026_flux_sweep` (entity `zhao-group`). One row per grid cell; each entry is a seed and its run id, linked to the run page. The 18 earlier runs in `flux_arms_gpu*.json` predate W&B logging and have no page.

Grid	Cell	Seed: run
arms	flux_anchored	101: u3iq97id 102: ot19w48c 103: g3iazwin 104: 4mx9l38a 105: gzs9e901 106: 1gav5w61 107: p8i6sp2v 108: qm9fddw2 109: c4i363am 110: kbc0h7bv
	flux_free	101: oyppbowh 102: 8h54lhih 103: hum6hv7i 104: rdqxfngq 105: gi56ouut 106: 3j67e7ou 107: 3v3j67wv 108: yepgwwki 109: 5q0zos2c 110: flkf5wzn
	flux_nullspace	101: lw7cdqt5 102: boqtw6j0 103: wgv3ogwk 104: 3x9sgbve 105: 87j1nvb4 106: dgojsxba 107: fay4r3sn 108: zw2up11b 109: e7cbxan6 110: m1h0dbd2
	flux_off	101: cvghzlru 102: 875pwhhp 103: 74ucn3q9 104: d6fufo9v 105: 90m94xzu 106: qx017bn6 107: 1epiwuli 108: 0klqi5gj 109: luid55f0 110: 29p85yl7
	pooled	101: czltmzp7 102: amie33pg 103: 7pxleywm 104: h9o1xexz 105: 71ftm5fd 106: 561svy7h 107: pq8nj79f 108: u4nh0znr 109: bw06snzd 110: 60ks7mhg
null	flux_anchored	301: avo7djgs 302: adcfdc1w 303: io7ugt8m 304: xy8srp0b 305: qv527rhn 306: up6z1xil 307: 29wn2436 308: 1bxevt3l 309: kb5r11yd 310: p7v9kj0l 311: hkfm29ji 312: 3c6k5fmd
	pooled	301: tb9lscvb 302: kkd93q06 303: n994wqto 304: 6aqns9oo 305: 03ew8rkh 306: 1hx74zr4 307: on67lnig 308: 7760u2hw 309: pi8kis29 310: 987qwj8g 311: 3djkhboa 312: cm1aem7j
reg	lr0.0001-wd0.0001-h32	201: m8kkt9sl 202: ocnkovj6
	lr0.0001-wd0.0001-h64	201: wdkjyjz6 202: 1ub1q45o
	lr0.0001-wd0.01-h32	201: soki18gv 202: 7914cdgv
	lr0.0001-wd0.01-h64	201: ptzdqpll 202: 1sebhjhx
	lr0.0003-wd0.0001-h32	201: vew8wglq 202: lskpseis
	lr0.0003-wd0.0001-h64	201: nux1vvp8 202: 57imkx5o
	lr0.0003-wd0.01-h32	201: 2x0g4fyx 202: ys9qw09i
	lr0.0003-wd0.01-h64	201: 9z732rhv 202: 3mix6sa1
	lr0.001-wd0.0001-h32	201: 0kru961j 202: 1xtp6xqd
	lr0.001-wd0.0001-h64	201: 24ku9gob 202: vt0deb5v
	lr0.001-wd0.01-h32	201: gigbzsy8 202: 3307ovbt
	lr0.001-wd0.01-h64	201: rtcriynx 202: ck53jtqk

The ordering looks like the ladder the design predicted. It is not evidence, for three measured reasons.

No arm clears its own null. A label-permutation null over 24 runs has peak mean 0.0962, standard deviation 0.0441, and a 95th percentile of **0.1545**. No arm reaches that percentile. The smallest p across all 22 configurations tested is 0.24, and the single highest peak anywhere in the sweep, 0.2110, belongs to a *null* run.

The last-5 column is the honest one. Every arm's peak sits far above its own end-of-training value, and three of five end negative. A maximum over 30 epochs of a statistic whose analytic null width at $n_{\text{val}} \approx 340$ is 0.0546 inflates that width empirically to 0.0962, a factor of 1.8.

The split moved with the seed. 026 re-rolled the train/validation partition per seed, so seed varied the nuisance and the signal together. Its paired arm gap has an across-seed standard deviation of 0.0303 with real labels and 0.0722 with permuted ones, both inflated by the partition moving underneath the contrast.

5.3 The defect that outranks all three ✗

Every betaxanthin run in this repository falls into one of two architecture families, and the family accounts for nearly all of the variance between runs.

Table 12: Measured validation Pearson by backbone. The best betaxanthin result ever obtained is 0.43399, GilaHyper job 1344, config `gh_metabolism_002`, seed 42, peak at epoch 39 of 80.

Family	Configuration	Measured
weak	2 graphs, learnable embeddings, MSE, hidden 32, 4 heads	0.04 to 0.16
strong	9 graphs, `prot_T5_all`, no learnable embedding, CRPS, hidden 90, 9 heads	0.32 to **0.43**

`train_flux.py` builds the **weak** family. So 026 compared a flux module against a CGT running roughly five times below its own demonstrated ceiling, and its `flux_off` arm at 0.0837 is the correct control *for that sweep* rather than a CGT baseline. Nothing in 026 tested whether the module helps a CGT that works, which is what the current run exists to do.

5.4 The baselines the module has to beat ✗

Table 13: Every measured comparator, with the statistic named. FCL is Merzbacher's Flux Cone Learning, a three-class classifier rather than a regressor, scored on its own 639-gene test split.

Comparator	Statistic	Value
CGT, strong backbone	max val Pearson, $n_{\text{val}} = 340$	0.43399
FCL `RandomForest_Resampled`	gene-level accuracy (majority 0.674)	0.700
FCL	Matthews correlation	0.232
FCL	Pearson against the raw screen	0.2435
FCL	Spearman against the raw screen	0.0391
fully-connected / MLP baseline	**not measured; does not exist**	–

There is no MLP baseline for betaxanthin anywhere in the repository. If one is wanted it must be built, and no number should be quoted for it until it is.

5.5 The noise floor, which sets the replicate count ✗

Four independent measurements of the across-seed standard deviation on this endpoint: 0.0302 from hyperparameter-matched reruns, 0.0355 from ten fresh seeds of `flux_off`, 0.0299 from ten seeds of `flux_anchored`, and 0.0337 from a three-seed 019 arm. The analytic single-correlation width at $n_{\text{val}} \approx 340$ is 0.0546.

Two consequences were computed rather than asserted. Selection inflation over 36 Optuna trials is 0.064, which puts the bias-corrected floor of the 0.4301 best at **0.3662**. And reaching an effective standard error of 0.010 needs **ten replicates per arm**, which is what sets the current run's twelve.

6 The grid on Delta ✗

6.1 What produced the 144 cells ✗

Job array `21813442_[1-8]` plus its replacement task `21814993_0` on `bfjt-delta-gpu`, partition `gpuA40x4`: nine nodes, four A40 per node, **two runs per GPU**, so 72 workers, each owning two cells and its own Optuna database. Six arms times 24 seeds is 144 cells, sharded by whole seed so that a wall-clock kill leaves every arm at the same seed count. Per-worker databases are what avoid the race that had several workers claiming one trial from a shared file and silently turning four cells into duplicates.

Two earlier attempts are part of the record. Array `21797666` on `bbub`, laid out as twelve seeds per arm, ran every worker into the embeddings coupling of Section 6.3 (`No gene embeddings available`) and completed seven tasks of six and a half hours each with zero epochs trained. Task 0 of `21813442` lost seven of its eight workers in the first minutes to `FileNotFoundError: data/sgd/genome/data.db`, which is the genome-rebuild race: the dataset constructor defaults to `overwrite=True`, so the first worker to open the genome deletes the file the other seven are reading. The fix is `overwrite=False`, which the older experiments already pass; tasks 1 to 8 read the corrected script at start and task 0 was canceled and resubmitted with fresh Optuna databases, since a canceled worker leaves its trial in `RUNNING` and a reused database would have dropped a cell without saying so.

The primary endpoint is Spearman on the pinned Merzbacher 639-gene test split at an epoch selected on validation, so the selection set and the reporting set are disjoint. The decision rule is the difference of differences in Section 4.

Table 14: Every environment fact confirmed on the node rather than assumed, since the repository's own files disagree about several of them.

Fact	Value
`DATA_ROOT`	`/scratch/bbub/mjvolk3/torchcell`. The Delta `.env` still says `/work/hdd`, which is wrong; pointing at a root with no build makes the dataset attempt a Neo4j rebuild on a compute node
interpreter	`/work/hdd/bbub/miniconda3/envs/torchcell/bin/python`, called directly, no `conda activate` and no container
GPU	NVIDIA A40, 46,068 MiB, four per node
`NUM_WORKERS`	0, forced. Spawned workers re-import off Lustre and stall
W&B	online, entity `zhao-group`, project `torchcell_027_bxfx`

6.2 Three faults found and fixed before submitting ✗

The Open Enzyme Database mirror was absent from Delta. `resolve_kcat_table` catches the resulting `FileNotFoundError` and returns an empty record list, which replaces every measured k_{cat} with the organism default. The enzyme-constrained arm would have run with no enzyme constraints and reported a plausible number. The mirror is 516 kB and was synced; the coverage gate exists to abort exactly this case, and it can only fire if there is a mirror to measure.

`WANDB_ENTITY` was unset, so runs would have landed under whatever default entity the node carried while 026's are under `zhao-group`, leaving the arms unjoinable at analysis time.

`OMP_NUM_THREADS` was unset, so eight workers on a 32-CPU allocation of a 64-core node would each have defaulted to the node's core count.

6.3 Three couplings the backbone change forced ✗

Moving 027 onto the strong backbone broke three things, each found by running a smoke test rather than by reading the code, and each silent in a different way.

1. The perturbation head defaults to eight attention heads and 90 is not divisible by eight, so its cross-attention refuses to build. 019 uses six.
2. The model reads `cell_graph["gene"].x` only when it is told it has precomputed embeddings. With the learnable table disabled and `node_embeddings` unset it finds neither and raises. It also indexes the embedding dataset to infer its width, so it needs the objects rather than their names.
3. Dropping 026's `build_dataset` dropped its module globals with it.

A two-epoch local smoke run on the corrected backbone moved loss 1.0586 to 1.0088 and validation Pearson -0.0214 to $+0.1055$, and computed the pinned test Spearman at $+0.0261$ on $n = 629$. Two epochs is not a result. What it establishes is that the nine-graph `prot_T5` backbone builds, trains and scores end to end.

6.4 The prespecified decision, on the full grid ✗

All eight array tasks of `21813442` completed on `bfjt-delta-gpu`, six to six and a half hours each, 64 workers, **no worker lost**, and the replacement task `21814993` completed in 6:30 with its own eight. 144 cells were written, giving **24 balanced seeds**, which is the number the design called for and more than twice the 10 the noise floor requires.

The analysis is `analyze_bxfx.py`, written before the grid launched and not edited after. It reports no maximum anywhere: the trainer selects the epoch on validation and scores on a disjoint test set, so every number is a mean over seeds with its across-seed standard error.

Table 15: Test Spearman on the pinned 639-gene Merzbacher split, at an epoch selected on validation, over the full 24 balanced seeds. The Flux Cone Learning RandomForest_Resampled baseline is the published value on the same 639 genes.

Arm	Mean	SD	SE
FCL RF (published)	+0.0391	–	–
flux_off (stoichiometry, no energies)	**+0.0388**	0.0541	0.0110
flux_free	+0.0294	0.0417	0.0085
pooled (no module)	+0.0269	0.0506	0.0103
null_pooled (permuted)	+0.0076	0.0298	0.0061
null_anchored (permuted)	−0.0017	0.0537	0.0110
flux_anchored (full module)	−0.0196	0.0544	0.0111

Table 16: Both criteria fail. D1 asks whether the paired arm gap exceeds the same gap with the labels destroyed; D2 asks whether the full module beats the published Flux Cone Learning baseline on the same genes.

Criterion	Value	Result
D1	excess over the permuted-label gap = -0.0372, 95% bootstrap CI $[-0.0724, -0.0031]$	**FAIL**
D2	flux_anchored -0.0196 ± 0.0111 against FCL RF $+0.0391$	**FAIL**

Verdict: the metabolic module is not shown to improve betaxanthin prediction.

Every one of the 144 cells is a W&B run with its per-epoch curves, and Table 17 links each by seed and arm, so any number above can be followed to the training it came from. The project is `zhao-group/torchcell_027_bxfx`.

Table 17: Every training run behind Tables 15 and 16, 144 in all, one per seed and arm. Each cell is the W&B run id, linked to the run page in project `torchcell_027_bxfx` (entity `zhao-group`), which holds the per-epoch training loss, validation loss, validation Pearson and the selected epoch.

Seed	pooled	flux_off	flux_free	flux_anchored	null_pooled	null_anchored
2700	8x6oiic5	rxaa1jqp	uews7le8	wsjte9kr	7aa0oazt	9ft3217z
2701	za1p0euu	k7bdmx0c	8fuby5nh	6wdet72a	uhh02hos	xa25q15s
2702	begcedhj	gpw2ddcl	w8r8k8vi	vr4t8xef	wbemdgt1	qn32rlpb
2703	rqydt43t	hznxj8mf	x0pbj45b	cyqo4dfs	a7g89vvx	v1kly7fl
2704	snv8hgv9	k6o32p3r	nqx5rj0d	yrklkmip	c5f52r4m	qh3bot4l
2705	giope3y3	pz2lordy	jrazf1w2	a30w9wqj	7xitbs73	7xfqqt4i
2706	0zai3k2o	dikzkekb	4a4q3f8u	5xidffpp	uamf2txt	88xk57iy
2707	95rjfqng	81ekfjmg	xsiqkeyj	oioaj7i3	v8cgxgxl	0luciagx
2708	59v554vc	wp105j57	ifevr877	qiutr4ut	de4bcbje	qz6pkjjb
2709	mav09tv8	1np2cfqm	fxcf4jz7	lacbpgws	6hkoxvfs	pj4evfkt
2710	cwbpa00w	binsbobp	nm1f8ait	3yoig7nj	wwoz8ar3	eo70l2cz
2711	ltij10df	8w5mp191	xmdz0udy	zj3avco2	xtx53e8c	e0fw796e
2712	e8qq2r1n	cq43i62s	aqzg9g4s	umim4rir	9q6qlw93	duahs4vx
2713	45kgbz4h	p4a4qjlq	ooy1aeep	nrjaasjb	14s3p794	y1cigozb
2714	mhzjpht8	33o585r6	3auf26jw	bimxb7sr	l7l8r41e	yndjhzp6
2715	h3irz6pe	szv1pyd1	8hqzdkq2	731g0frn	wdbhnjn5	15lutdk9
2716	eoo2dvad	6z9vh6le	4vmfl47g	ageh81z9	h56o8g95	oh0x0swg
2717	fahajjn1	q4dfso6d	b10ecc2e	w7v7cxdi	vdgwczs2	kc734t2l
2718	trlnwz1v	af9vvwpe	gvowlp9w	te3t2svi	ljnwivnr	u3gzp530
2719	bmyxtq7a	otcxl9pt	dl47veg8	tvi275q8	us70savs	jw7r3c3t
2720	gt4v9lhn	fu06w3xw	bujbzgpm	8gzhglaf	r2bhzfst	1g716ke9
2721	ubllusbn	ya55ia5u	o13lpzyw	xve395ii	8a10vkgw	mlxb61ps
2722	2fx6zu6z	4ogrdcz3	8hszg7jv	kx308jej	pni6hr7c	1m9bffbv
2723	e8i19jo7	d3blys8c	edumtzy4	ymt5m7mw	nqo9mnla	4nr0mfzs

The D1 interval moved twice, and the sequence is worth recording. At the first eight balanced seeds it was $[-0.1172, -0.0237]$, excluding zero. At 20 seeds it was $[-0.0787, +0.0026]$, including zero, and the claim that the module was *measurably worse* was withdrawn on that basis. At the full 24 it is $[-0.0724, -0.0031]$ and excludes zero again, by 0.0031. **The prespecified sample size was 24, so that is the result; the two earlier readings were interim looks at a design that had not finished.** The verdict was FAIL at every one of the three, which is the part that did not depend on where the interval landed. The honest summary of the magnitude is that the anchored module is worse than no module by an amount the full design can just resolve from zero, and anyone reading -0.0031 as a comfortable margin has read it wrong.

The permuted arms are why D1 is not just the raw gap. With the training labels destroyed the same contrast is -0.0093 ± 0.0124. So roughly a fifth of the raw -0.0465 gap is reproduced by an architecture that cannot have learned anything, which is exactly the confound D1 exists to subtract and exactly what a comparison against zero would have credited to biology.

Stoichiometry alone is the one arm that does not hurt. `flux_off`, which keeps the network and the box but switches off the energies, is the top measured arm at $+0.0388$ and sits level with the published Flux Cone Learning baseline at $+0.0391$, a difference far inside one standard error and not a claim of either direction. Its contrast against no module at all is $+0.0118 \pm 0.0144$, indistinguishable from zero. **The cost is specifically in the thermodynamic anchor**: `flux_anchored` minus `flux_free` is -0.0490 ± 0.0135, so learning the potential beats reading the tabulated one.

6.5 Correction: every cell is a constant predictor ✗

The numbers above were read as a comparison of trained models. An audit of the prediction dumps the trainer writes at the selected epoch, `audit_prediction_collapse.py`, says they are not. **Every one of the 144 cells outputs a near-constant across the 905 held-out genotypes.** The largest prediction spread in any cell is 0.8 percent of the target spread, and the median across cells is under 10^{-5} of it. Pearson and Spearman are scale-free, so a correlation can still be computed from the residual jitter, and that is what every entry in Tables 15 and 16 is: the correlation of jitter five orders of magnitude below the signal with the truth.

Table 18: Spread of the predictions across the 905 held-out genotypes at the selected epoch, per arm over its 24 seeds. The targets have standard deviation 0.623 and mean +0.068 in standardized units. SD ratio is the prediction standard deviation over the target standard deviation; a value near zero is a constant predictor. Constant is the range, over seeds, of the value the model outputs for every genotype.

Arm	Cells	SD ratio, max	SD ratio, median	Constant, over seeds
`pooled`	24	8.3e-6	6.7e-7	−0.06 to −0.00
`flux_off`	24	8.1e-3	1.4e-6	−4.33 to +4.37
`flux_free`	24	4.9e-3	8.6e-7	−6.64 to +4.68
`flux_anchored`	24	5.1e-3	3.8e-6	−7.08 to +2.91
`null_pooled`	24	5.2e-6	2.3e-7	−0.08 to −0.00
`null_anchored`	24	3.1e-3	2.6e-6	−2.19 to +3.92

The two families of collapse are different, and the difference explains the arm ordering without any biology. The no-module arms sit at the target mean, which is the loss-optimal constant under CRPS when the model has found no signal; their training loss reads 1.00 from epoch 10 onward, the standardized variance, and never moves again. The flux arms sit at constants as far as −7 from the mean, so they have not even fit the intercept: their training loss is still falling at epoch 199 (12 percent over the last 50 epochs for `flux_anchored`), driven by the penalty terms rather than the data term. A model whose output is a wrong constant scores below a model whose output is the right constant on any correlation computed from jitter, and that is the whole of the "cost of the thermodynamic anchor" reported above.

What survives. The verdict that the module is *not shown* to improve betaxanthin prediction stands, because nothing in the grid predicts betaxanthin. The stronger reading, that the anchored module is measurably worse and that the cost sits in the tabulated energies, is withdrawn: a comparison between two collapsed models says nothing about either. The D1 interval excluding zero by 0.0031 is a property of the jitter.

Where the defect must be. The same nine-graph `prot_T5` backbone reached validation Pearson 0.434 on this target under the 019 Lightning trainer (job 1344, Table 12), so the target is learnable and the architecture can learn it. The 027 loop in `train_bx.py` is a hand-written trainer sharing none of that code path. Hypothesis, untested: the pooled arm's immediate settle at the mean, within ten epochs and at every seed, points at the loss or the inputs rather than at optimization noise, and the first things to check are whether the perturbation actually reaches the model in this loop, whether the CRPS head's $\log\sigma$ is absorbing the fit, and the standardization of the masked pooled target. The trainer also logs no prediction spread, which is the one diagnostic that would have caught this at epoch 10 rather than after ten node-days; `pred_sd_ratio` goes next to `val_loss` in the next version.

What this does and does not settle. Twenty-four seeds, one phenotype, and a trainer that did not learn in any arm. **None of the parameter tables built in this work was wired in**, and after the correction above the grid tests neither the module's structure nor its parameters. The order of work is now fixed: first a trainer that reproduces the 0.434 on the no-module arm, then the module arms on that trainer, and only then the merged $\Delta_f G$ table and a predicted k_{cat} table.

6.6 The plateau, and the recipe that leaves it ✗

The constants are not a defect in the loop. Twenty-nine single-GPU runs on GilaHyper on 2026-09-09 (slurm jobs 1610 to 1638, one script and one log per run under `experiments/027-betaxanthin-metabolic/`) show that every betaxanthin run, on every trainer and every commit since July, starts on the same mean plateau: a validation prediction spread near 10^{-4} of the target spread and a training loss of exactly the standardized variance. What varies is when a run leaves it, and that is set by the recipe. The 019 trainer on today's tree with the July recipe, betaxanthin alone, a Gaussian CRPS head and the graph regularizer at 6.5×10^{-5} over two layers, has a spread ratio of 0.003 at epoch 16, 0.12 at epoch 23 and 0.42 at epoch 28, with validation Pearson 0.20, 0.26, 0.32 and 0.39 at epochs 23, 24, 27 and 28 and 0.25 to 0.40 through epoch 34 (job 1635). Five commits from 2026-07-26 to 07-31 re-run the same way escape at epochs 21 to 24 (jobs 1623, 1626, 1630, 1632). The August 020 grid, with hard attention masking at six layers and no regularizer, escaped at epochs 117 to 157 in its W&B histories, which is a slower recipe rather than a slower code base.

The grid above ran neither element. Its loop sliced $\log\sigma$ off the CRPS head before the loss, so the head was scored by MSE, and it never passed the regularizer to the model, so `graph_reg_loss` was zero; under that recipe 200 epochs were not enough for any of the 144 cells. Both are now switches on `train_bx.py`, which also logs the spread ratio every epoch; under CRPS plus regularizer its own pooled cell leaves the plateau on the same path as the 019 trainer at a slower clock: spread ratio 0.021 at epoch 16, 0.12 at epochs 26 to 28, 0.30 to 0.40 at epochs 32 to 35, with validation Pearson 0.15 to 0.28 over epochs 27 to 35 (job 1634). A ten-epoch collapse abort, which the first diagnostics ran with, would have killed every one of these runs; the abort's patience now defaults to 200. The regularizer alone had not lifted the spread by epoch 21 (job 1638), where the combined recipe was already at 0.025, so CRPS is doing at least part of the work. The module arms under the combined recipe, `flux_off` and the new `flux_capacity` arm with a predicted

per-gene enzyme abundance in the capacity bound (jobs 1636, 1637), were still on the plateau at epoch 31 with a spread ratio under 10^{-3} and their constraint terms dominating the loss at 3 to 6 against a data term near 0.5. The flux arms escape later than the pooled arm, if they escape at this budget at all, and that is the next measurement: the same recipe, a longer budget, and the spread ratio read before any correlation.